

Graduate Microeconomics

Marshallian Demand and Indirect Utility Function

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Outline

- 1 Marshallian Demand
- 2 Changes in Income
- 3 Changes in Prices
- 4 The Indirect Utility Function
- 5 Roy's Identity

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Marshallian Demand

Given prices $p \in \mathbb{R}_{++}^L$ and income $y > 0$, define

$$B(p, y) = \{x \in \mathbb{R}_+^L : p \cdot x \leq y\}.$$

The demand correspondence

$$\mathcal{D}(p, y) = \arg \max_{x \in B(p, y)} u(x).$$

It reports the **set of optimal bundles** at each (p, y) .

If the solution is unique, write $\mathcal{D}(p, y) = \{x(p, y)\}$. The vector $x(p, y)$ is the **Marshallian demand function**.

Also called *Walrasian* or *ordinary* demand. Each $x_i(p, y)$ depends on the entire price vector and income.

Demand Is Homogeneous

Degree zero in prices and income

Proposition

For every $t > 0$,

$$\mathcal{D}(tp, ty) = \mathcal{D}(p, y).$$

If demand is unique, $x(tp, ty) = x(p, y)$.

Proof. The two budget sets are identical:

$$tp \cdot x \leq ty \iff p \cdot x \leq y.$$

The consumer ranks the same feasible bundles using the same preferences. Changing the monetary unit changes neither opportunities nor choices. No convexity or nonsatiation assumption is needed for this argument.

Relative Prices and Real Income

Normalize prices and income together

Homogeneity lets us choose good L as the numeraire:

$$x(p, y) = x\left(\frac{p_1}{p_L}, \dots, \frac{p_{L-1}}{p_L}, 1, \frac{y}{p_L}\right).$$

Divide **all prices and income** by the same positive number.

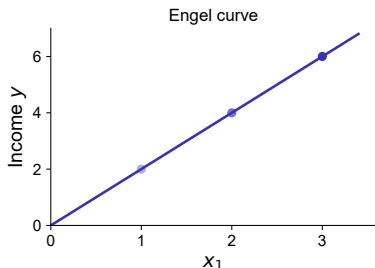
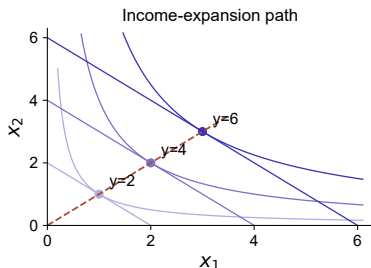
- Doubling all prices and income leaves demand unchanged.
- Doubling prices alone reduces purchasing power.
- Doubling income alone changes the budget set.

Homogeneity in (p, y) holds generally. Proportionality in *income alone* will require homothetic preferences.

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Income Shifts the Budget

Fix p and vary y . Budget lines shift in parallel.



Special case: $u(x) = \sqrt{x_1 x_2}$ and $p = (1, 1)$. Homothetic preferences give a ray from the origin; in general, the path can curve.
The income-expansion path records entire bundles; the Engel curve records one good.

Engel Curves Trace Income Effects

An **Engel curve** traces $x_i(\bar{p}, y)$ at fixed prices $\bar{p}$. Our graphs put quantity horizontally and income vertically. Where demand is differentiable, good i is locally

$$\begin{array}{ll} \text{normal} & \frac{\partial x_i}{\partial y} > 0, \\ \text{inferior} & \frac{\partial x_i}{\partial y} < 0. \end{array}$$

A zero derivative means no local income effect.

- The classification concerns demand at given prices and income.
- A good can be normal at low income and inferior at high income.
- “Inferior” is a response to income, not a judgment of quality.

Income Elasticities

At $x_i > 0$, define the income elasticity and budget share:

$$\eta_i = \frac{\partial x_i}{\partial y} \frac{y}{x_i}, \quad s_i = \frac{p_i x_i}{y}.$$

- Inferior: $\eta_i < 0$.
- Normal necessity: $0 < \eta_i < 1$.
- Unit income elasticity: $\eta_i = 1$.
- Luxury: $\eta_i > 1$.

At fixed prices,

$$\frac{\partial \log s_i}{\partial \log y} = \eta_i - 1.$$

A luxury's spending share rises with income; a necessity's share falls.

Income Effects Must Add Up

Engel aggregation

Differentiate Walras' law with respect to income:

$$\sum_{i=1}^L p_i x_i(p, y) = y \quad \Longrightarrow \quad \boxed{\sum_{i=1}^L p_i \frac{\partial x_i}{\partial y} = 1.}$$

When the elasticities are defined, substitute $\partial x_i / \partial y = \eta_i x_i / y$:

$$\boxed{\sum_{i=1}^L s_i \eta_i = 1.}$$

- All goods cannot be inferior at the same point.
- With positive demands, all goods cannot be strict necessities.
- Nor can all goods be luxuries: the weighted average is exactly one.

Homothetic Demand Scales with Income

With homothetic preferences and a unique optimum,

$$x(p, ty) = t x(p, y), \quad t > 0.$$

Proof by rescaling feasible choices.

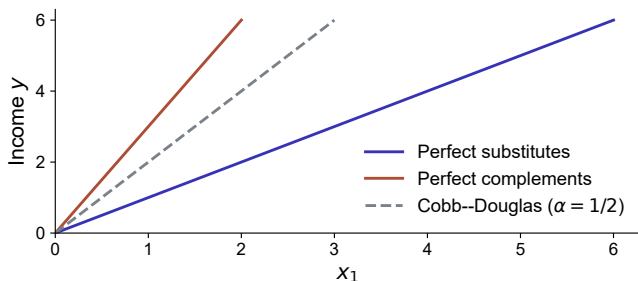
- 1 z is feasible at income ty exactly when z/t is feasible at income y .
- 2 Optimality at y gives $x(p, y) \succeq z/t$.
- 3 Homotheticity gives $tx(p, y) \succeq z$.

Thus $tx(p, y)$ is optimal at ty ; uniqueness identifies it as demand. Every **positively demanded** good has $\eta_i = 1$. A good with zero demand can remain unconsumed; its elasticity is undefined.

Homothetic Engel Curves Are Linear

Three benchmark preferences

At $p_1 = 1$, $p_2 = 2$, each benchmark is homothetic.



Straight Engel curves through the origin need not have the same slope. Homotheticity restricts income responses, not the way demand responds to relative prices.

Income Independence Is Conditional

Quasi-linear demand and the numeraire constraint

Consider $u(x) = \log(1 + x_1) + x_2$ with $p = (1/2, 1)$. Its demand is

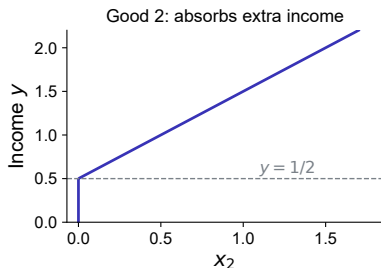
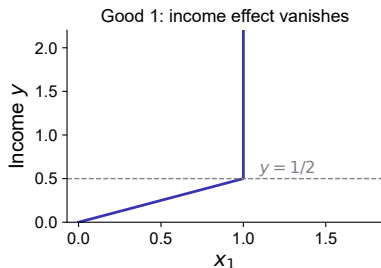
$$x_1(p, y) = \min\{2y, 1\}, \quad x_2(p, y) = \max\{y - 1/2, 0\}.$$

- If $0 < y < 1/2$, the consumer buys only good 1: $\partial x_1 / \partial y = 2$.
- If $y > 1/2$, good 2 is consumed positively: $\partial x_1 / \partial y = 0$ and $\partial x_2 / \partial y = 1$.
- At $y = 1/2$, demand is continuous but not differentiable in income.

Income independence for the nonlinear good applies where the numeraire is consumed positively.

Income Independence Is Conditional

The Engel curves



Above the threshold, good 1's Engel curve is **vertical** because quantity is on the horizontal axis. The extra income is spent on good 2. Walras' law continues to hold on both sides of the threshold.

Two Patterns of Income Response

Homothetic and quasi-linear preferences

These structures imply very different income responses.

Homothetic preferences

$$x_i(p, y) = y x_i(p, 1).$$

- Unit elasticity for positive demands.
- Shares do not vary with income at fixed prices.
- The whole bundle scales.

Quasi-linear preferences

With positive numeraire demand,

$$\frac{\partial x_i}{\partial y} = 0$$

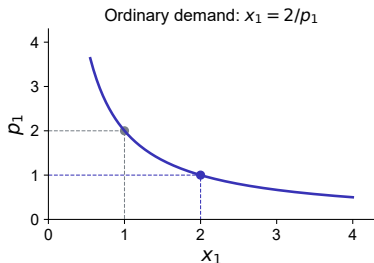
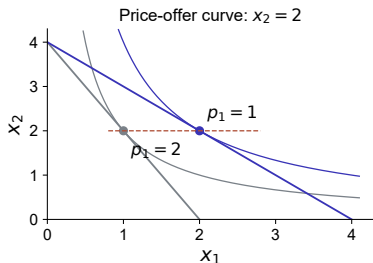
for nonlinear goods.

- Zero income elasticity for these positive demands.
- The numeraire absorbs income changes.

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Prices Rotate the Budget

Fix income and all other prices; vary p_i .



Example: $u(x) = \sqrt{x_1 x_2}$, $y = 4$, $p_2 = 1$. A price-offer curve lives in commodity space; an own-demand curve plots price against quantity.

Ordinary and Giffen Goods

At a differentiable point, good i is locally

$$\begin{array}{ll} \text{ordinary} & \frac{\partial x_i}{\partial p_i} < 0, \\ \text{Giffen} & \frac{\partial x_i}{\partial p_i} > 0. \end{array}$$

These derivatives hold **money income** and all other prices fixed.

- A price change alters both relative prices and purchasing power.
- Utility maximization alone does not force every ordinary demand curve to slope downward.
- “Normal” concerns income; “ordinary” concerns own price.

Separating the two channels will require compensated demand in a later topic.

Own-Price Elasticity

For $x_i > 0$, define the **signed** elasticity

$$\varepsilon_{ii} = \frac{\partial x_i}{\partial p_i} \frac{p_i}{x_i}.$$

For an ordinary good:

$$\begin{aligned} -1 < \varepsilon_{ii} < 0 & \quad \text{inelastic,} \\ \varepsilon_{ii} = -1 & \quad \text{unit elastic,} \\ \varepsilon_{ii} < -1 & \quad \text{elastic.} \end{aligned}$$

The effect on expenditure is

$$\frac{\partial(p_i x_i)}{\partial p_i} = x_i(1 + \varepsilon_{ii}).$$

Spending rises with price for an inelastic demand and falls for an elastic demand.

Cross-Price Effects Are Gross

For $j \neq i$, holding income and other prices fixed:

$$\frac{\partial x_i}{\partial p_j} \begin{cases} > 0 & : \text{good } i \text{ is a gross substitute for } j, \\ < 0 & : \text{good } i \text{ is a gross complement to } j. \end{cases}$$

- The response includes purchasing-power effects.
- Gross cross-price effects need not be symmetric: $\partial x_i / \partial p_j \neq \partial x_j / \partial p_i$ in general.
- Cobb–Douglas: $\partial x_1 / \partial p_2 = 0$.
- Perfect complements: $\partial x_1 / \partial p_2 = -y / (p_1 + p_2)^2 < 0$.

A zero gross cross-price effect is not a general statement about the absence of substitution.

Homogeneity Restricts Elasticities

Suppose x_i is differentiable. Differentiate $x_i(tp, ty) = x_i(p, y)$ with respect to t at $t = 1$:

$$\sum_{j=1}^L p_j \frac{\partial x_i}{\partial p_j} + y \frac{\partial x_i}{\partial y} = 0.$$

If $x_i > 0$, divide by x_i :

$$\sum_{j=1}^L \varepsilon_{ij} + \eta_i = 0.$$

A common 1% increase in all prices and income has zero total effect on demand. For Cobb–Douglas: $\varepsilon_{ii} = -1$, $\varepsilon_{ij} = 0$ for $j \neq i$, and $\eta_i = 1$.

Price Effects Must Add Up

Differentiate the budget identity

Differentiate $\sum_i p_i x_i(p, y) = y$ with respect to p_j :

$$\underbrace{x_j}_{\text{direct cost effect}} + \underbrace{\sum_{i=1}^L p_i \frac{\partial x_i}{\partial p_j}}_{\text{adjustment of quantities}} = 0.$$

With positive demands, multiply by p_j/y to obtain

$$\sum_{i=1}^L s_i \varepsilon_{ij} = -s_j.$$

The direct cost of a price increase must be offset by changes in quantities, since total expenditure remains equal to income.

Demand-System Restrictions

At a differentiable point with positive demands:

Source	Elasticity restriction
Walras' law, income derivative	$\sum_i s_i \eta_i = 1$
Homogeneity, for each good i	$\sum_j \varepsilon_{ij} + \eta_i = 0$
Walras' law, price derivative	$\sum_i s_i \varepsilon_{ij} = -s_j$

These are **necessary restrictions**; arbitrary regression coefficients need not satisfy them. They are not, by themselves, sufficient to establish that a demand system comes from utility maximization.

Elasticities Must Be Consistent

A numerical check

Two goods have budget shares $s_1 = 0.4$ and $s_2 = 0.6$. Suppose the estimated elasticities are

$$\eta = (1.5, 1), \quad E = \begin{pmatrix} -1.2 & 0.1 \\ 0.2 & -0.8 \end{pmatrix}, \quad E_{ij} = \varepsilon_{ij}.$$

Can these describe a differentiable, budget-exhausting, homogeneous demand system?

- Income restriction: $0.4(1.5) + 0.6(1) = 1.2 \neq 1$.
- Homogeneity, row 1: $-1.2 + 0.1 + 1.5 = 0.4 \neq 0$.
- Price restriction, column 1: $0.4(-1.2) + 0.6(0.2) = -0.36 \neq -0.4$.

Each discrepancy is a distinct way to detect a violation of the restrictions.

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The Indirect Utility Function

Definition

$$v(p, y) = \max_{x \in B(p, y)} u(x).$$

The indirect utility function reports the **maximum utility value**.

Object	Inputs	Output
$u(x)$	Consumption bundle	Utility
$\mathcal{D}(p, y)$	Prices and income	Optimal bundles (argmax)
$v(p, y)$	Prices and income	Best utility (max)

Even if demand is multivalued, v is a single number: all optimal bundles have the same utility.

Demand Determines Attained Utility

Example: Cobb–Douglas

For $u(x) = x_1^\alpha x_2^{1-\alpha}$, recall

$$x_1(p, y) = \frac{\alpha y}{p_1}, \quad x_2(p, y) = \frac{(1 - \alpha)y}{p_2}.$$

Evaluate utility at this bundle:

$$\begin{aligned} v(p, y) &= \left(\frac{\alpha y}{p_1} \right)^\alpha \left(\frac{(1 - \alpha)y}{p_2} \right)^{1-\alpha} \\ &= \underbrace{\alpha^\alpha (1 - \alpha)^{1-\alpha}}_{A > 0} y p_1^{-\alpha} p_2^{-(1-\alpha)}. \end{aligned}$$

The indirect utility function summarizes optimal welfare at every (p, y) . No new optimization problem is being solved here.

Properties of Indirect Utility

With continuous u , $X = \mathbb{R}_+^L$, $p \gg 0$, $y > 0$:

- 1 v is homogeneous of degree zero in (p, y) .
- 2 v is non-increasing in each price.
- 3 v is non-decreasing in income; under LNS, strictly increasing.
- 4 v is continuous in (p, y) .
- 5 v is quasiconvex jointly in (p, y) .

These value-function properties do not require a unique optimum. Convex preferences concern mixtures of consumption bundles. Quasiconvex indirect utility concerns mixtures of *price–income pairs*.

Nominal Scaling Preserves Utility

Homogeneity of degree zero

For any $t > 0$, $B(tp, ty) = B(p, y)$. Therefore

$$v(tp, ty) = \max_{x \in B(tp, ty)} u(x) = \max_{x \in B(p, y)} u(x) = v(p, y).$$

This is the same budget-set argument used for Marshallian demand. If v is differentiable, Euler's identity gives

$$\sum_i p_i v_{p_i}(p, y) + y v_y(p, y) = 0.$$

A proportional change in all nominal magnitudes has no effect on attainable utility.

More Income Raises Utility

If $y' > y$, then $B(p, y) \subset B(p, y')$. Hence

$$v(p, y') \geq v(p, y).$$

Why is the inequality strict under LNS?

- 1 Let $x \in \mathcal{D}(p, y)$. At the higher income, it has budget slack.
- 2 A sufficiently small neighborhood of x in X remains affordable.
- 3 LNS supplies a strictly preferred affordable bundle z .

Thus $v(p, y') \geq u(z) > u(x) = v(p, y)$. Strictly increasing does not alone imply a strictly positive derivative at every point. Roy's differentiable formula will require $v_y > 0$ at the point considered.

Higher Prices Cannot Raise Utility

If $p' \geq p$ componentwise, then $B(p', y) \subseteq B(p, y)$. Therefore

$$v(p', y) \leq v(p, y).$$

The inequality need not be strict when only one price rises.

An unconsumed good

For $u(x) = x_1 + x_2$, $p_1 > p_2$ implies

$$x = (0, y/p_2), \quad v(p, y) = y/p_2.$$

Increasing p_1 further has no effect on utility.

Once Roy's identity applies, the local condition is explicit:

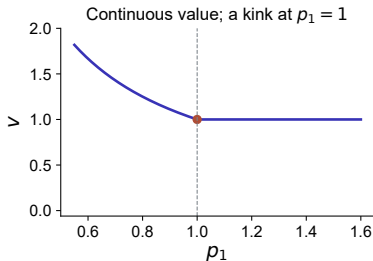
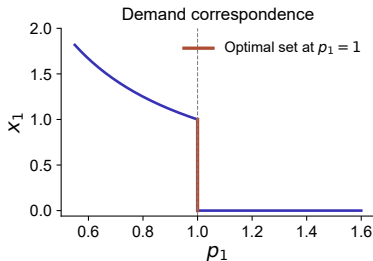
$$v_{p_i} = -v_y x_i.$$

Indirect Utility Is Continuous

Example: a switch between optimal choices

For $u(x) = x_1 + x_2$ with $p_2 = y = 1$,

$$v(p_1, 1, 1) = \max\{1/p_1, 1\}.$$



As p_1 crosses 1, demand switches corners, but the two alternatives give the same utility at the switch. Value remains continuous.

Indirect Utility Is Continuous

The maximum theorem

On the positive-price, positive-income domain:

- The budget set varies continuously with (p, y) and is compact.
- Utility is continuous in the consumption bundle.
- The maximum theorem therefore gives continuous $v(p, y)$.

The conclusion concerns the **best attainable value**, even when multiple bundles tie for that value.

Continuity alone says nothing about whether derivatives exist. The perfect-substitutes example has a continuous value function with a kink.

Indirect Utility Is Quasiconvex

Proposition

For every utility level $\bar{v}$, the lower contour set

$$\{(p, y) : v(p, y) \leq \bar{v}\}$$

is convex.

Equivalently, for $t \in [0, 1]$,

$$v(tp + (1 - t)p', ty + (1 - t)y') \leq \max\{v(p, y), v(p', y')\}.$$

If two price–income situations deliver utility at most $\bar{v}$, averaging their prices and incomes cannot deliver utility above $\bar{v}$. This statement follows from [linear budgets](#), not convex preferences.

Indirect Utility Is Quasiconvex

Proof: compare the budget sets

Set $p^t = tp + (1 - t)p'$ and $y^t = ty + (1 - t)y'$, with $0 < t < 1$. Take any $z \in B(p^t, y^t)$. If z were unaffordable in *both* original budgets, then

$$p \cdot z > y, \quad p' \cdot z > y'.$$

Weight and add these inequalities:

$$p^t \cdot z > y^t,$$

contradicting affordability at the averaged budget. Therefore

$$B(p^t, y^t) \subseteq B(p, y) \cup B(p', y').$$

The inclusion is in the **union**, not necessarily the intersection.

Indirect Utility Is Quasiconvex

Proof: compare the optimal values

Every bundle affordable at (p^t, y^t) is affordable in at least one original budget. Hence, for each such bundle z ,

$$u(z) \leq \max\{v(p, y), v(p', y')\}.$$

Maximize over the averaged budget:

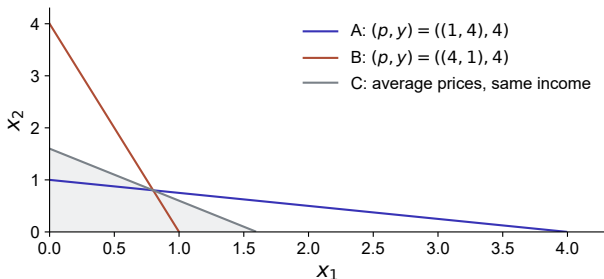
$$v(p^t, y^t) \leq \max\{v(p, y), v(p', y')\}.$$

- The endpoint utilities need not be equal.
- The argument does not differentiate utility or demand.
- Quasiconvexity is weaker than convexity.

Mixing *prices and income* is a different operation from mixing consumption bundles.

Indirect Utility Is Quasiconvex

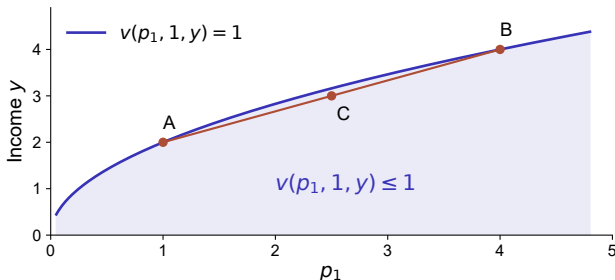
Example: the averaged budget



For $u(x) = \sqrt{x_1 x_2}$, both endpoint budgets give $v = 1$. The averaged prices $(2.5, 2.5)$ at income 4 give $v = 0.8$. The shaded budget lies inside the union of the endpoint budgets; its best bundle cannot outperform both endpoint optima.

Indirect Utility Is Quasiconvex

Example: the lower contour set



With $v(p_1, 1, y) = y/(2\sqrt{p_1})$, $v \leq 1$ means $y \leq 2\sqrt{p_1}$. Points $A = (1, 2)$ and $B = (4, 4)$ have utility 1. Their midpoint $C = (2.5, 3)$ has $v = 3/\sqrt{10} < 1$.

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Roy's Identity Recovers Demand

Roy's identity

Under the standard consumer-theory and local regularity conditions, if v is differentiable and $v_y(p, y) > 0$,

$$x_i(p, y) = - \frac{v_{p_i}(p, y)}{v_y(p, y)}.$$

- Numerator: utility loss from a marginal price increase.
- Denominator: utility gained from a marginal income increase.
- Their ratio recovers the quantity demanded.

The formula needs a nonzero denominator. LNS and continuity alone are not differentiability assumptions.

Roy's Identity Recovers Demand

Proof: the envelope theorem

Recall the consumer's Lagrangian, with nonnegativity multipliers included:

$$\mathcal{L} = u(x) + \lambda(y - p \cdot x) + \mu \cdot x.$$

Apply the envelope result from the previous topic at a regular optimum:

$$v_{p_i}(p, y) = \mathcal{L}_{p_i} = -\lambda x_i(p, y), \quad v_y(p, y) = \mathcal{L}_y = \lambda.$$

If $\lambda > 0$, divide the two expressions:

$$-\frac{v_{p_i}}{v_y} = -\frac{-\lambda x_i}{\lambda} = x_i.$$

We use the envelope theorem; we do not solve the first-order conditions again.

Roy's Identity Recovers Demand

Proof: keep the chosen bundle affordable

Fix $x^* \in \mathcal{D}(p, y)$. Under LNS, $p \cdot x^* = y$. At any nearby price q , give the consumer income $q \cdot x^*$. Since x^* is affordable,

$$F(q) := v(q, q \cdot x^*) \geq u(x^*) = F(p).$$

Thus F has a minimum at $q = p$. If v is differentiable,

$$0 = \left. \frac{\partial F}{\partial q_i} \right|_{q=p} = v_{p_i}(p, y) + v_y(p, y)x_i^*.$$

Dividing by $v_y > 0$ gives Roy's identity again. This argument does not differentiate demand and does not require an interior consumption bundle.

Roy's Identity Recovers Demand

Example: Cobb–Douglas

Recall $v(p, y) = Ay p_1^{-\alpha} p_2^{-(1-\alpha)}$. Its derivatives have a convenient form:

$$v_{p_1} = -\frac{\alpha v}{p_1}, \quad v_{p_2} = -\frac{(1-\alpha)v}{p_2}, \quad v_y = \frac{v}{y}.$$

Roy's identity gives

$$x_1 = -\frac{-\alpha v/p_1}{v/y} = \frac{\alpha y}{p_1}, \quad x_2 = \frac{(1-\alpha)y}{p_2}.$$

The common scale factor cancels. We recover the same demand without re-solving the UMP.

Roy's Identity Allows Corners

Differentiability is the key condition

With perfect substitutes and $p_1 < p_2$,

$$v(p, y) = y/p_1, \quad v_{p_1} = -y/p_1^2, \quad v_{p_2} = 0, \quad v_y = 1/p_1.$$

Roy correctly gives the corner bundle

$$x_1 = y/p_1, \quad x_2 = 0.$$

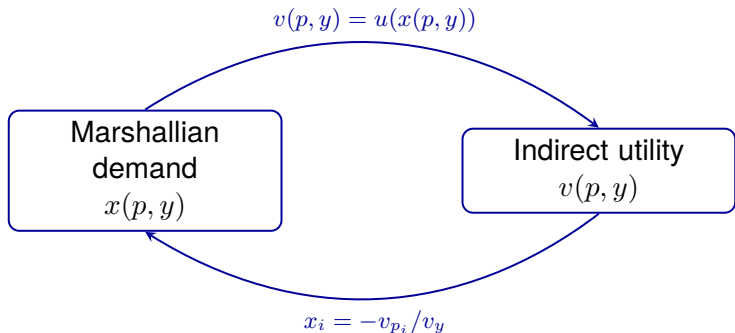
At $p_1 = p_2$, however,

$$v(p, y) = \max\{y/p_1, y/p_2\}$$

is not differentiable in prices and demand is multivalued. A

corner does not itself invalidate Roy. The differentiability and positive-denominator conditions matter.

Demand and Value Are Linked



- Demand records optimal behavior; indirect utility records the optimal value.
- Budget geometry gives homogeneity, monotonicity, and quasiconvexity.
- Roy's identity translates marginal welfare changes into demand.